

Survey Analysis and Requirements

Requirements to Use Case Diagram Generator

1 SURVEY OVERVIEW & DATA BASELINE

To understand the real needs of users, we conducted a requirements survey with 13 undergraduate software engineering students working on academic projects. The survey covered current diagramming habits, comfort with UML notations, major difficulties faced, prioritized features, input and output preferences, and privacy expectations. Below is the brief analysis of the collected survey data.

2 KEY DATA COLLECTION FINDINGS

- **Low UML Comfort Level:** 69.2% of respondents reported low or no comfort with standard UML and DFD notations, showing a clear need for automated guidance.
- **Time-Consuming Manual Work:** 76.9% identified manual diagram drafting as excessively time-consuming, taking focus away from actual development.
- **Difficulty in Translation:** 46.2% find it hard to translate written requirements into correct diagram elements (actors, use cases, and boundaries).
- **Top Priority Diagram Type:** Use Case diagram emerged as the absolute top priority (76.9% top priority, 84.6% overall support), followed by Activity diagrams (46.2%) and Flow diagrams (46.2%).
- **Preferred Requirement Formats:** 69.2% prefer uploading an SRS in PDF format, 53.8% functional requirements, 46.2% Word documents (DOCX), and 46.2% user stories.
- **Interactive Editing Demand:** 84.6% want to add or remove elements on an interactive canvas, and 76.9% need drag-and-drop repositioning after generation.
- **Ambiguity Clarification:** 76.9% want the tool to ask for clarification when requirements are unclear, and 53.8% want problematic lines highlighted.
- **Turnaround & Privacy:** Over 53.8% expect generation within 30 to 60 seconds. 53.8% strictly require that uploaded files are not stored permanently on servers.

3 KEY INSIGHTS GAINED FROM SURVEY

- **Editable Drafts over Static Images:** Students do not want a black-box tool that gives a final locked image. They need a flexible draft that they can edit, customize, and refine.
- **Pre-Generation Confirmation:** 69.2% want to preview detected actors and actions before the diagram is drawn, preventing AI interpretation errors early.
- **Handling Changing Requirements:** 38.5% struggle with requirements churn. When requirements change, 61.5% want the tool to ask for confirmation and then update related diagrams.
- **Transparency & Assumptions:** 61.5% want a clear list of assumptions made by the tool so that students can defend their design during project evaluation.
- **Strict Session Privacy:** Students and developers do not want their project documents permanently saved. Temporary in-memory processing is essential.

4 FUNCTIONAL REQUIREMENTS

- FR-1 Document Ingestion:** The system shall allow users to upload SRS documents in PDF and DOCX formats, as well as paste user stories and plain text.
- FR-2 Actor and Entity Preview:** The system shall extract actors, system boundaries, and actions from the requirements and display them for user review prior to diagram rendering.
- FR-3 Automated Use Case Synthesis:** The system shall automatically generate standard UML Use Case diagrams showing primary actors, system boundary boxes, and use case bubbles.
- FR-4 Interactive Canvas Editor:** The system shall provide an interactive visual canvas allowing users to move elements, add/delete use cases, rename labels, and reconnect associations.
- FR-5 Ambiguity Detection & Clarification:** The system shall identify ambiguous, missing, or contradictory requirements and highlight them with a prompt for user clarification.
- FR-6 Traceability & Assumption Logging:** The system shall map each generated use case back to the original requirement line and output an explicit list of assumptions made.
- FR-7 Multi-Format Export:** The system shall allow users to export diagrams in PDF, PNG, SVG, and text-based PlantUML formats.

5 NON-FUNCTIONAL REQUIREMENTS

NFR-1 **Generation Speed:** The system shall process a standard requirements document and generate the initial diagram within 30 to 60 seconds.

NFR-2 **Data Privacy & Auto-Deletion:** Uploaded files and extracted text shall not be permanently stored. All session data shall be automatically deleted immediately after diagram generation.

NFR-3 **Usability & Learning Curve:** The user interface shall be simple and intuitive, allowing any student to generate diagrams without requiring prior tool training.

NFR-4 **UML 2.5 Compliance:** Generated diagrams shall strictly follow standard UML 2.5 notation rules (correct stick figure actors, horizontal ellipses, dashed include/extend arrows).

NFR-5 **System Reliability:** The system shall provide fallback error handling and clear explanations whenever an input document is corrupted or unparseable.

6 PROJECT SCOPE & ROADMAP SUMMARY

Based on the empirical findings, the initial version of the project (MVP) will focus squarely on automated Use Case diagram generation from PDF and text requirements, paired with an interactive visual editor and instant PDF/PNG export. Advanced features, such as Activity diagram generation and automatic Git synchronization, will be rolled out in subsequent phases.